discussed in section 11.4.6. When used for client authentication, there are two modes that can be used:

- In self-signed certificate authentication, the client registers a certificate with the AS that is signed by its own private key and not by a CA. The client authenticates to the token endpoint with its client certificate and the AS checks that it exactly matches the certificate stored on the client's profile. To allow the certificate to be updated, the AS can retrieve the certificate as the `x5c` claim on a JWK from a HTTPS URL registered for the client.
- In the PKI (public key infrastructure) method, the AS establishes trust in the client's certificate through one or more trusted CA certificates. This allows the client's certificate to be issued and re-issued independently without needing to update the AS. The client identity is matched to the certificate either through the Subject DN or SAN fields in the certificate.

Unlike JWT bearer authentication, there is no way to use mTLS to obtain an access token for a service account, but a client can get an access token using the client credentials grant. For example, the following curl command can be used to obtain an access token from an AS that supports mTLS client authentication:

```
curl -d 'grant_type=client_credentials&scope=a+b+c' \
    -d 'client_id=test' \    #A
    --cert test-client.pem \    #B
    --key test-client-key.pem \    #B
    https://as.example.org/oauth2/access_token
```